

JAYDEN PIAO

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EDUCATION

University of British Columbia — BSc Computer Science (GPA: 3.9/4.33)

Graduating May 2026

Relevant Coursework: Distributed Systems, AI/ML, Databases, Data Structures, Operating Systems, Parallel Computation

EXPERIENCE

Google (Incoming Summer 2026)

May 2026 – July 2026

Software Engineer Intern

Sunnyvale, CA

- Cloud SQL - Replication and Availability

Amazon Web Services (AWS)

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, WA

- Built a Rust physical replication system for DynamoDB's new LSM-tree storage engine, replacing the logical approach
- Reduced replica recovery time by 60% and CPU utilization by 50% by streaming byte-level data directly over custom netcat pipelines using Unix domain sockets and TCP bridges
- Fulfills Paxos protocol, enabling cross-host replica synchronization and quorum recovery with Linux keyring re-encryption

UBC MINT (Multifaceted Innovations in Neurotechnology)

Mar 2024 – Aug 2025

Software Engineering Lead

Vancouver, BC

- Led backend development in Rust for a Brain-Computer Interface (BCI), simulating EEG signal processing for real-time neural data pipelines
- Engineered Rust backend for high-frequency time-series EEG data, with simulated signal generation to emulate real-world asynchronous input, aligned with scientific computing workflows

Marr Labs (YC W24)

May 2024 – Dec 2024

Software Engineer Intern

Remote

- Designed and implemented evaluation frameworks for AI agents (text and voice), applying principles of observability and automation to enable rapid iteration and data-driven development
- Collaborated cross-functionally to integrate RESTful tools and visualization features that surfaced performance regressions and reduced manual testing by hours per engineer
- Developed scalable systems using Docker, FastAPI, and AWS ECS - contributing to team-wide agent adoption and iterative ML workflows

The Verse

May 2023 – Aug 2023

Software Engineer Intern

Remote

- Developed 'U4Ea Labels,' a MERN stack application that performs sentimental music analysis and uses DALL-E to generate custom artwork in the form of nutritional labels, serving 37K+ customers with personalized designs
- Built a MERN stack internal management tool with OAuth authentication, featuring employee accounts to track activity and engagement, integrated with a custom Discord bot built with JavaScript for automated monitoring

UBC Thunderbots

Sept 2021 – Aug 2022

Software Engineer

Vancouver, BC

- Developed real-time robotics control systems using C++, emphasizing low-latency data processing and performance tuning on Ubuntu systems
- Engineered Python-based referee leveraging TensorFlow ML models which improved real-time decision-making by 30%

PROJECTS

Quorum Coordination Service | [Source Code](#)

Go | Multi-Paxos | TLA+ | HTTP | WAL/Snapshots

- Built a 3-node Multi-Paxos coordination service with linearizable reads, WAL+snapshot recovery, leases, and watch streams; added fault injection and a batched-write path improving burst throughput by 5.3x under stability testing.

Hey Spotify | [Source Code](#)

Python | FastAPI | JavaScript | SQL

- Building full-stack Spotify voice assistant with OAuth PKCE auth and schema-validated intent execution pipeline
- Added exponential backoff for API rate limits, request-ID structured JSON logs, and p95 latency instrumentation (<2s)

PhoneticAI | [Source Code](#) | [DevPost](#)

Python | JavaScript | React | MySQL

- Built a platform for AI agents with tailored personalities to handle calls, using LLMs for information retrieval, scheduling, and analytics; integrated real-time SIP call routing and webhook-based event tracking via Twilio and Retell AI

TECHNICAL SKILLS

Programming Languages: Python, Rust, Go, SQL, JavaScript/TypeScript, Java, C/C++, HTML, CSS

Technologies: PyTorch, TensorFlow, React, FastAPI, Node.js, PostgreSQL, Firebase, Tailwind CSS, Git, AWS